

# Factors Associated with Antibiotic Use for Acute Bronchitis

Ralph Gonzales, MD, MSPH, Paul H. Barrett, Jr., MD, MSPH, Lori A. Crane, PhD, MPH,  
John F. Steiner, MD, MPH

**OBJECTIVES:** To describe the clinical features of adults diagnosed with acute bronchitis, and to identify clinical variables associated with antibiotic treatment of acute bronchitis.

**DESIGN:** Prospective, cohort study.

**SETTING:** Primary care office practices at a group-model HMO in the Denver metropolitan area.

**PATIENTS/PARTICIPANTS:** Patients were adults seeking care for acute respiratory illnesses. Participating clinicians included internists, family medicine physicians, nurse practitioners, physician assistants, and registered nurses.

**MEASUREMENTS AND MAIN RESULTS:** Clinicians voluntarily completed encounter forms for patients presenting with acute respiratory illnesses between February and May, 1996. Acute bronchitis was the primary diagnosis in 16% of acute respiratory illness visits ( $n = 1,525$ ). The most frequent symptoms of acute bronchitis were cough (92%), phlegm production (63%), "runny nose" (50%), and throat pain (50%). The most frequent physical examination findings were pharyngeal erythema (45%), cervical lymphadenopathy (19%), wheezes (18%), and rhonchi (17%). Antibiotics were prescribed to 85% of patients diagnosed with acute bronchitis. Purulent nasal discharge by patient report, and sinus tenderness on physical examination were moderately associated with antibiotic treatment ( $p = .06$  and  $.08$ , respectively). Antibiotic prescription rates did not vary by patient age or gender, duration of illness, days of work lost due to illness, or clinician type.

**CONCLUSIONS:** Acute bronchitis is frequently treated with antibiotics in ambulatory practice. The clinical factors we identified to be associated with antibiotic use for acute bronchitis appear to play a minor role in explaining the excessive use of antibiotics for this condition. These findings suggest that clinicians use the diagnosis of acute bronchitis as an indication for antibiotic treatment, despite clinical trials and expert recommendations to the contrary.

**KEY WORDS:** physician practice patterns; acute respiratory illness; antibiotic treatment; acute bronchitis.

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Address correspondence and reprint requests to Dr. Gonzales: University of Colorado Health Sciences Center, Division of General Internal Medicine, Box B-180, 4200 East Ninth Ave., Denver, CO 80262.

The rapidly rising prevalence of penicillin-resistant *Streptococcus pneumoniae* (PRSP) in our communities has underscored the urgency for reducing unnecessary antibiotic use in ambulatory practice.<sup>1-3</sup> Studies of the epidemiology of PRSP have consistently identified a strong association between prior antibiotic use and the risk of acquiring PRSP.<sup>1,4-6</sup> In light of these findings, the Centers for Disease Control and Prevention and other authorities have intensified their recommendations and efforts to promote judicious antibiotic use.<sup>7-10</sup> A recent study of physician practice patterns in the United States identified bronchitis as an important target for reducing unnecessary antibiotic use in ambulatory practice.<sup>11</sup>

Antibiotics are prescribed in 60% to 90% of cases of acute bronchitis in the United States and abroad.<sup>11-16</sup> The practice of prescribing antibiotics for acute bronchitis persists despite microbiologic evidence supporting a viral etiology in the vast majority of cases.<sup>13,15,17</sup> Indeed, numerous randomized, controlled trials have demonstrated no meaningful clinical benefit from antibiotic therapy for most patients without underlying lung disease, and antibiotic therapy is not recommended.<sup>18-28</sup>

Understanding why antibiotic use persists under these circumstances requires a better understanding of the clinical encounter for patients with acute bronchitis. Studies describing the clinical features of acute bronchitis have been limited to either patient populations fulfilling predefined case definitions, or retrospective chart reviews in which the documentation of symptoms and signs had not been standardized.<sup>15,18-24,29</sup> Variation in the clinical presentation of these patients might explain why some patients are treated with antibiotics and some are not.

The aims of the present study are to describe the clinical features of patients diagnosed with acute bronchitis, and to identify clinical variables associated with antibiotic use.

## METHODS

### Study Design

We conducted a prospective, cohort study of patients presenting to Kaiser Permanente, Colorado Region medical facilities with acute respiratory illness.

### Study Setting

Kaiser Permanente of Colorado is a group-model HMO serving about 250,000 adult members in the Denver-Boulder metropolitan area. Members received medical care services solely from Kaiser Permanente clinicians,

staff, laboratory, and pharmacy. Study sites included the three largest Kaiser Permanente primary care facilities located within the Denver Metropolitan area. As a group, these three sites accounted for approximately one third of all ambulatory care visits to Kaiser Permanente. Most patients with acute or urgent illnesses were given same-day appointments in these practices, rather than being seen in urgent care centers or emergency departments.

## Study Subjects

*Eligible patients* included consecutive adults presenting for care at one of the three study facilities between February and May (1996) with acute respiratory illness, defined as recent onset of symptoms referable to the chest, throat, or nasal passages. Patients aged less than 18 years were excluded. *Eligible clinicians* included internal medicine and family practice board-certified physicians, nurse practitioners, physician assistants, and registered nurses.

## Data Collection

In order to standardize the documentation of duration of illness, past medical history, symptoms, and physical examination findings present at the office visit, a patient encounter form, the Acute Respiratory Illness (ARI) form, was designed to replace the traditional free-text medical record note. This form was designed with significant input from physicians and nurses who would use the form. Nurses checking-in the patient completed the sections on history of current illness, symptoms, past medical history, current medications, and medication allergies. The clinician completed the sections for physical examination findings, assessment, and plan. The form was printed as a carbonless duplicate, which allowed for a copy of each form to be forwarded to the investigators on a weekly basis, while the original copy remained a part of the patient's medical record. The ultimate decision to use the form was left to the discretion of the clinician. Data from the ARI forms were merged with data from the Kaiser Permanente administrative database, which included, for each office visit, principal diagnosis, as coded by the *International Classification of Diseases, 9th Revision, Clinical Modification* (ICD-9-CM), patient characteristics (age, gender), and clinician information, as well as pharmacy and diagnostic test utilization.

## Acute Bronchitis Study Cohort

We compiled 1,525 ARI forms during the 4-month study period. From this cohort, we analyzed those visits with a diagnosis of acute bronchitis on either the ARI form or the Kaiser Permanente database. Office visits were included in our analysis if (1) a diagnosis of acute bronchitis was written on the encounter form (medical chart), or (2) an encounter form was completed, no writ-

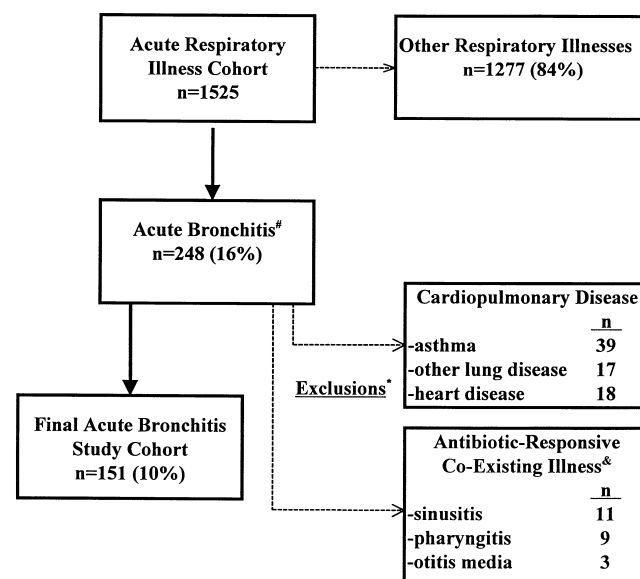
ten diagnosis was identifiable, and the visit was coded as acute bronchitis (ICD-9-CM code 466.0) in the diagnostic database. In order to evaluate the impact of clinical features of acute bronchitis on the decision to treat with antibiotics, we sought to eliminate cases of acute bronchitis for which antibiotic therapy might be justified. Therefore, we excluded patients in whom a past history of cardiopulmonary disease was documented, or an antibiotic-responsive illness coexisted, e.g., sinusitis. A flow diagram depicting the assembly of the study cohort is shown in Figure 1.

## Statistical Methods

We performed  $\chi^2$  tests with appropriate degrees of freedom, or Fisher's Exact Test (when subgrouping resulted in no more than five office visits per group), to compare the frequency of clinical characteristics between patients treated and not treated with antibiotics.

## RESULTS

Acute bronchitis was diagnosed in 16% of acute respiratory illnesses for which adults sought medical attention during our study period. Of these bronchitis cases, 24% involved underlying cardiopulmonary disease, and 15% had a coexisting diagnosis for which antibiotics are sometimes indicated; these two groups were not mutually exclusive (Fig. 1). Our final study cohort consisted of 151 clinical encounters for patients with acute bronchitis, who received care from 64 different clinicians. The median



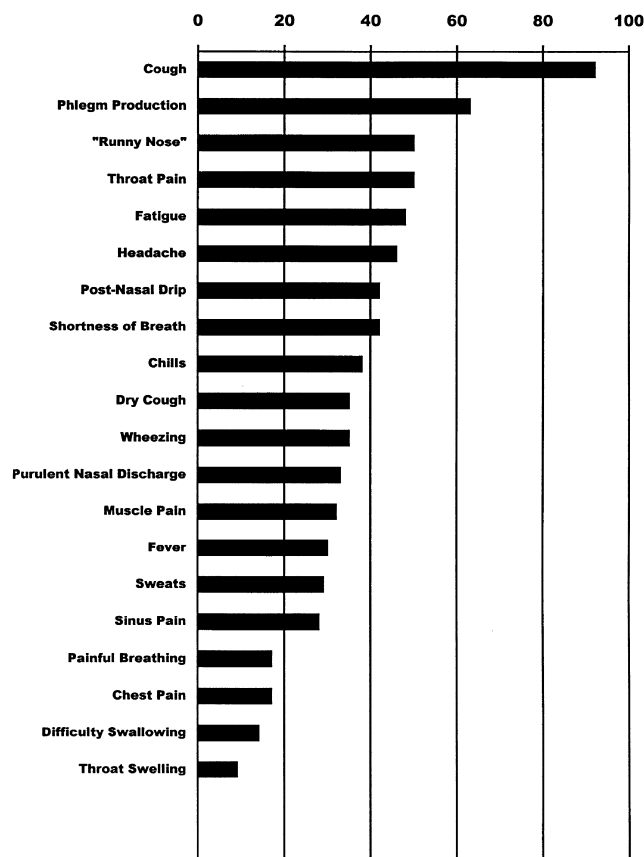
**FIGURE 1.** Patient selection for acute bronchitis study cohort. \*Unique visits diagnosed with acute bronchitis; \*Some visits had more than one exclusion condition; &Antibiotic-responsive coexisting illness present at the same office visit. Only the three most common illnesses are shown.

number of case patients seen per clinician in our study cohort was two, and the range was one to nine.

Table 1 displays the patient and clinician characteristics of the 151 adults diagnosed with acute bronchitis after excluding those with cardiopulmonary disease and other antibiotic-responsive coexisting illnesses. Females, nonsmokers, and patients 18 to 44 years of age predominated. Sixty percent of patients sought care within 1 week of onset of symptoms, and 79% within 2 weeks. At least 1 day of missed work due to illness occurred in 43% of patients. Physicians directly provided care to 34% of the patient visits, whereas 50% of the visits were attended by a registered nurse, and the remaining visits were attended by a nurse practitioner or physician assistant. The medical record and prescriptions resulting from registered nurse visits require physician oversight and cosignature.

Figure 2 shows the frequency of symptoms reported by patients diagnosed with acute bronchitis. The most common symptom was cough (92%), followed in frequency by phlegm production (63%), "runny nose" (50%), throat pain (50%), and fatigue (48%). Other common pulmonary symptoms included shortness of breath (42%), wheezing (35%), and painful breathing (17%).

In contrast to the range and frequency of symptoms reported, few physical examination findings were present at the office visit (Fig. 3). Pharyngeal erythema was most



**FIGURE 2.** Symptoms reported by patients diagnosed with acute bronchitis ( $n = 151$ ). Patients were asked to report on the presence of "symptoms you have experienced within the last 2 days due to your illness."

**Table 1. Patient and Clinician Characteristics in Clinical Encounters for Acute Bronchitis**

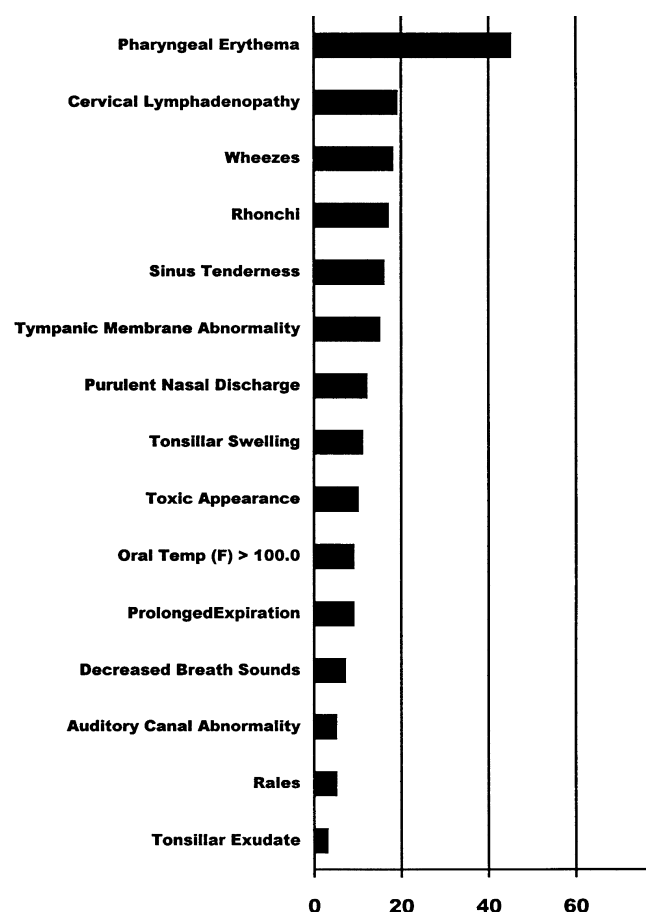
| Characteristic*                                                | %  |
|----------------------------------------------------------------|----|
| <b>Patient characteristics</b>                                 |    |
| Age, years ( $n = 150$ )                                       |    |
| 18–44                                                          | 55 |
| 45–64                                                          | 33 |
| $\geq 65$                                                      | 12 |
| Gender ( $n = 150$ )                                           |    |
| Male                                                           | 41 |
| Female                                                         | 59 |
| Tobacco use ( $n = 151$ )                                      | 30 |
| <b>Illness characteristics</b>                                 |    |
| Duration of illness prior to seeking care, weeks ( $n = 141$ ) |    |
| $\leq 1$                                                       | 60 |
| 1–2                                                            | 19 |
| $> 2$                                                          | 21 |
| Missed work due to illness, days ( $n = 105$ )                 |    |
| 0                                                              | 57 |
| 1                                                              | 21 |
| 2                                                              | 14 |
| $\geq 3$                                                       | 8  |
| <b>Clinician type (<math>n = 151</math>)</b>                   |    |
| Physician                                                      | 34 |
| Nurse practitioner, physician's assistant                      | 17 |
| Registered nurse, licensed practical nurse                     | 50 |

\*Sample sizes for each characteristic vary owing to nonresponse (duration of illness, missed work days), and missing information from merged administrative databases (age, gender).

common (45%), followed by cervical lymphadenopathy (19%), wheezes (18%), rhonchi (17%), and sinus tenderness (16%). Chest radiography was the most common diagnostic test used in this population, ordered in 11% of cases.

The most common prescriptions for patients with acute bronchitis were antibiotics and bronchodilators, prescribed to 85% and 17% of patients, respectively. We evaluated whether the variation in sociodemographic factors, duration of illness and missed work, specific symptoms reported, physical examination findings present, or clinician type correlated with antibiotic treatment (outlined in Table 1 and Figs. 2 and 3). As shown in Table 2, purulent nasal discharge by patient report, and sinus tenderness on physical examination were present more frequently in patients treated with antibiotics than in those not treated ( $p = .06$  and  $.08$ , respectively). The remaining factors showed no significant association with antibiotic treatment, including provider type, bronchodilator therapy and chest x-ray utilization.

Because clinicians might differ in how they weight certain symptoms or physical examination findings for specific competing diagnoses (e.g., sinusitis, pharyngitis,



**FIGURE 3.** Physical examination findings in patients diagnosed with acute bronchitis ( $n = 151$ ). Due to nonresponse,  $n = 149$  for oral temperature measurement. Clinicians recorded the presence or absence of each finding for each patient.

or pneumonia), we also analyzed whether the presence of at least one symptom or examination finding for a competing diagnosis was associated with antibiotic use. The presence of at least one symptom of sinusitis (purulent nasal discharge, sinus pain, or postnasal drip) differed between patients treated with antibiotics and those not treated (69% and 48%, respectively,  $p = .05$ ) (Table 2). However, there was no significant difference between groups based on at least one physical examination finding of sinusitis, at least one symptom or examination finding of pharyngitis, or at least one lung sound abnormality.

## DISCUSSION

Acute bronchitis is a common condition for which adults seek care in the United States. For the past 20 years, bronchitis has consistently ranked among the 10 most frequent ambulatory care diagnoses recorded in the National Ambulatory Medical Care Surveys conducted by the National Center for Health Statistics.<sup>30-34</sup> During this same period, multiple randomized, controlled trials, as well as evidence-based reviews of these trials, have dem-

onstrated little or no meaningful clinical benefit for antibiotic treatment of acute bronchitis in most healthy adults.<sup>18-27</sup> In spite of these reports, studies of physician prescribing practices for acute bronchitis in the United States reveal that antibiotics continue to be prescribed at high rates for this self-limited condition.<sup>11,12</sup>

Studies to date have been unable to delineate the reasons why physicians continue to prescribe antibiotics for acute bronchitis. We recently analyzed data from the 1992 National Ambulatory Medical Care Survey, and found that antibiotic prescribing does not vary substantially by patient sociodemographic factors, payment sources, practice location, or physician specialties.<sup>11</sup> The present study was conducted to evaluate whether information routinely gathered in the clinical encounter for acute bronchitis is influencing this physician behavior.

In our cohort, antibiotics were prescribed in 85% of cases of acute bronchitis, after excluding patients with underlying lung disease and antibiotic-responsive coexisting illnesses. This rate is similar to antibiotic prescription rates reported previously in the United States and abroad (which have ranged from 66% to 100%).<sup>11,12,14-16</sup>

Several factors could explain this high rate of antibiotic use for acute bronchitis despite the paucity of evidence supporting it. Because acute bronchitis is a clinical diagnosis for patients presenting with a new cough, one possible explanation for prescribing antibiotics for this condition is clinician uncertainty regarding the presence of pneumonia. However, evaluation of a clinical prediction rule for pulmonary infiltrates in adult patients with acute respiratory illness found that cough, sputum production, and dyspnea were equally common in patients with and without pneumonia.<sup>35</sup>

Although not measured in the present study, variation in patient expectations and demands for antibiotics could contribute to the high rates of antibiotic use for acute bronchitis. Some patients seek care for acute respiratory illnesses, such as upper respiratory infections, with the explicit intent of getting a prescription for antibiotics,<sup>36-38</sup> and clinicians appear to prescribe antibiotics in response to these expectations.<sup>38,39</sup>

A final possible explanation for the high antibiotic prescription rate for acute bronchitis is that clinicians believe antibiotics benefit acute bronchitis, despite the evidence in the medical literature. When we reviewed the management of acute bronchitis in several popular general medicine and family medicine textbooks, we found a wide range of treatment recommendations that both promoted<sup>40,41</sup> and denounced<sup>42,43</sup> antibiotic use for acute bronchitis. Infectious disease texts and experts, however, were consistent in not recommending antibiotic therapy.<sup>44,45</sup>

Misclassification, resulting from improper assignment of diagnosis or antibiotic prescription, threatens the validity of our findings. Because we were interested in describing the features of patients diagnosed by clinicians as having acute bronchitis, we did not seek to impose a case definition for acute bronchitis. In order to include

the broadest possible spectrum of patients with acute bronchitis in our study, we assembled our cohort based on either a written diagnosis on the encounter form, or an ICD-9 coded diagnosis in the administrative database when a written diagnosis was not present on the encounter form. It is possible that diagnosis coding sheets, by only including a selected number of check-off diagnoses, encouraged clinicians to select acute bronchitis rather than an unlisted diagnosis. However, since the major competing diagnoses with acute bronchitis (illnesses such as pneumonia, upper respiratory infection, and asthma) are also

listed on the sheet, this misclassification should be minimized. Comparing patients with acute bronchitis selected by these two different mechanisms revealed few differences in symptoms and physical exam findings, and no difference in antibiotic prescription rates (data not shown).

Misclassification of antibiotic prescription could occur if patients were given antibiotic samples, and not written a prescription. This event would lead to underestimating the true antibiotic prescription rate. Because of the high prescription rate recorded in our study, and the lack of incentive to provide antibiotic samples to this HMO population

**Table 2. The Association Between Clinical Features of Acute Bronchitis and Antibiotic Use**

| Patient, Illness, and Clinician Characteristics           | Antibiotic Treatment, %<br>(n = 128) | No Antibiotic Treatment, %<br>(n = 23) | p Value |
|-----------------------------------------------------------|--------------------------------------|----------------------------------------|---------|
| Patient age, years (n = 150)                              |                                      |                                        | NS      |
| 18-44                                                     | 56                                   | 52                                     |         |
| 45-64                                                     | 33                                   | 30                                     |         |
| ≥65                                                       | 11                                   | 17                                     |         |
| Female gender (n = 150)                                   | 58                                   | 65                                     | NS      |
| Smoker (n = 151)                                          | 30                                   | 30                                     | NS      |
| Duration of illness prior to seeking care, days (n = 141) |                                      |                                        | NS      |
| 0-7                                                       | 57                                   | 74                                     |         |
| 8-14                                                      | 21                                   | 5                                      |         |
| >14                                                       | 21                                   | 21                                     |         |
| Missed work due to illness, days (n = 105)                |                                      |                                        | NS      |
| 0                                                         | 55                                   | 69                                     |         |
| 1                                                         | 21                                   | 23                                     |         |
| ≥2                                                        | 24                                   | 8                                      |         |
| Clinician type (n = 151)                                  |                                      |                                        | NS      |
| Physician                                                 | 34                                   | 35                                     |         |
| Nurse practitioner or physician assistant                 | 17                                   | 13                                     |         |
| Registered nurse                                          | 49                                   | 52                                     |         |
| Symptoms (n = 151)                                        |                                      |                                        |         |
| Headache                                                  | 45                                   | 57                                     | NS      |
| Fever                                                     | 31                                   | 22                                     | NS      |
| Sweats                                                    | 31                                   | 17                                     | NS      |
| Chills                                                    | 39                                   | 35                                     | NS      |
| Muscle pain                                               | 32                                   | 35                                     | NS      |
| Fatigue                                                   | 49                                   | 43                                     | NS      |
| Sinus pain                                                | 28                                   | 30                                     | NS      |
| Postnasal drip                                            | 43                                   | 39                                     | NS      |
| Runny nose                                                | 51                                   | 48                                     | NS      |
| Purulent nasal discharge                                  | 36                                   | 17                                     | .06     |
| Throat pain                                               | 49                                   | 57                                     | NS      |
| Throat swelling                                           | 9                                    | 9                                      | NS      |
| Difficulty swallowing                                     | 13                                   | 17                                     | NS      |
| Cough                                                     | 91                                   | 96                                     | NS      |
| Yellow phlegm                                             | 35                                   | 48                                     | NS      |
| Green phlegm                                              | 31                                   | 22                                     | NS      |
| Wheezing                                                  | 34                                   | 39                                     | NS      |
| Shortness of breath                                       | 43                                   | 39                                     | NS      |
| Painful breathing                                         | 16                                   | 22                                     | NS      |
| Chest pain                                                | 17                                   | 13                                     | NS      |

(continued)

(90% of whom have drug benefits), it is unlikely that unrecorded antibiotic use was of any significance.

Potential threats to the generalizability of our study include selection bias, surveillance bias, and restricting our study to a managed care patient and clinician population. Our study reduced the potential for selection and surveillance bias by deriving our clinical encounter data from a large sample of clinicians ( $n = 64$ ), thereby ensuring that the clinical information and prescribing practices measured in our study were not skewed by a few "outlier" clinicians. Because clinicians were not aware of the study question when completing encounter forms for "acute respiratory illness", potential surveillance bias affecting the recording of signs and symptoms was also reduced.

The generalizability of our findings depends on whether our cohort is a representative sample of adults diagnosed with acute bronchitis outside the Kaiser Permanente health care system. In order to compare our findings with previous work in this area, we searched for studies that prospectively recorded symptoms and signs from unselected patients in a standardized manner. We identified only two studies which fulfilled these criteria: the Medical Research Council and Public Health Laboratory Service collaborative study (MRC/PHLS), and a study by Boldy and colleagues.<sup>15,17</sup> These studies differed from the present study, however, in prompting physicians to use a pre-determined

case definition (MRC/PHLS), and in the acquisition of clinical data by the study investigators rather than by practicing clinicians. In all three studies, the most frequent symptoms reported were similar. Differences in study populations included an older population in the MRC/PHLS study, and a greater proportion of smokers in the Boldy study. These differences may account for the greater prevalence of wheezing and productive cough reported by patients in these studies, since advanced age and smoking history both increase the likelihood of underlying lung disease.

Unfortunately, the MRC/PHLS and Boldy studies did not publish complete reports of the physical examination findings present in their populations. We compared our physical examination findings with those reported by randomized control trials (RCTs) evaluating antibiotic therapy for acute bronchitis.<sup>18-25</sup> Although patients in RCTs more often reported productive cough (often an entry criteria for the RCTs), the frequency of other physical examination findings (pharyngeal erythema, abnormal lung exam, wheezing) in our sample was similar to those reported in RCTs.

Our study demonstrates that antibiotics continue to be prescribed at very high rates for acute bronchitis, and that clinicians do not appear to base their treatment decision on patient history or physical examination findings.

Table 2. (Continued)

| Patient, Illness, and Clinician Characteristics                                                       | Antibiotic Treatment, %<br>( $n = 128$ ) | No Antibiotic Treatment, %<br>( $n = 23$ ) | <i>p</i> Value |
|-------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------|----------------|
| Physical examination findings ( $n = 151$ )                                                           |                                          |                                            |                |
| Oral temp $>100^{\circ}\text{C}$                                                                      | 9                                        | 5                                          | NS             |
| Toxic appearance                                                                                      | 11                                       | 4                                          | NS             |
| Tympanic membrane abnormality                                                                         | 16                                       | 9                                          | NS             |
| Auditory canal abnormality                                                                            | 5                                        | 0                                          | NS             |
| Purulent nasal discharge                                                                              | 13                                       | 4                                          | NS             |
| Sinus tenderness                                                                                      | 18                                       | 4                                          | .08            |
| Pharyngeal erythema                                                                                   | 43                                       | 39                                         | NS             |
| Tonsillar swelling                                                                                    | 13                                       | 4                                          | NS             |
| Tonsillar exudate                                                                                     | 2                                        | 4                                          | NS             |
| Cervical lymphadenopathy                                                                              | 20                                       | 13                                         | NS             |
| Prolonged expiration                                                                                  | 8                                        | 17                                         | NS             |
| Decreased breath sounds                                                                               | 5                                        | 13                                         | NS             |
| Rales                                                                                                 | 5                                        | 4                                          | NS             |
| Rhonchi                                                                                               | 16                                       | 22                                         | NS             |
| Wheezes                                                                                               | 17                                       | 22                                         | NS             |
| Any sinusitis symptoms<br>(postnasal discharge, purulent nasal discharge, or sinus pain)              | 69                                       | 48                                         | .05            |
| Any pharyngitis symptoms<br>(throat pain, throat swelling, or difficulty swallowing)                  | 50                                       | 61                                         | NS             |
| Any sinusitis exam findings<br>(purulent nasal discharge or sinus tenderness)                         | 25                                       | 9                                          | .09            |
| Any pharyngitis exam findings<br>(tonsillar swelling, tonsillar exudate, or cervical lymphadenopathy) | 29                                       | 17                                         | NS             |
| Any lung sound abnormality<br>(rales, rhonchi, wheezes, or decreased breath sounds)                   | 66                                       | 57                                         | NS             |

These practice patterns suggest clinicians use the diagnosis of acute bronchitis as an indication for antibiotic treatment. Efforts to reduce antibiotic use for acute bronchitis should start with educating clinicians that acute bronchitis, in most cases, does not benefit from antibiotic therapy. Further research should explore how patient expectations and demands influence clinician prescribing practices for acute bronchitis, and how individual clinician prescribing profiles vary within and between practices.

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## African Drums

African drums, a slow, steady beat . . .  
faster, faster, faster  
STOP.  
Silence.

Should I move to the rhythm  
or lie still as I have been told to do?  
The drums again, slow, steady,  
faster, faster, faster  
STOP.  
Silence.

Are they outside dancing,  
dressed in brightly colored hues and rich textures,  
with feathers of eagles and painted faces?  
Perhaps a healing ritual  
or a form of prayer and  
sacred celebration?

The drums again, slow, steady . . .  
the sound of metal magnets  
circling, attracting, repelling,  
taking snapshots of my brain.

Faster, faster, faster  
Stop.  
Silence.

The intercom clicks on  
The MRI technician's voice fills the cold room . . .  
Hold your breath, don't move.

There is no dance here and there are no feathers.  
The costumes are all white.

I close my eyes and  
the drums begin again.  
Slow, steady  
faster, faster, faster  
STOP.  
Silence.

I pray for the healing ritual to begin.

ALICIA M. CONILL, MD  
*University of Pennsylvania, Philadelphia*  
FINALIST  
*1998 Creative Medical Writing Contest*